

Economic Growth TD2

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Exercise 1: Using a production function within Solow

An economy described by the Solow growth model has the following production function:

$$y(t) = \sqrt{k(t)}$$

where y denotes output per worker, $y(t) = \frac{Y(t)}{L(t)}$, and $k(t)$ denotes capital per worker, $k(t) = \frac{K(t)}{L(t)}$

1. Solve for the steady state value of y as a function of parameters s , n and δ (where the notation is the same as the model developed in class). Hint: don't forget to derive the fundamental differential equation.
2. A developed country has a saving rate of 28% and a population growth rate of 1% per year. A less developed country has a saving rate of 10% and a population growth rate of 4% per year. In both countries, $\delta=0.04$. Find the steady-state value of y for each country.
3. What policies might the less developed country pursue to raise its level of income?

Exercise 2: Introduction exercise (Midterm 2021)

Consider the following production function:

$$Y(t) = K(t)^\alpha L(t)^{1-\alpha}$$

where $0 < \alpha < 1$. Population $L(t)$ grows at the constant rate $n > 0$. The saving rate is constant and equal to s , with $0 < s < 1$.

1. Derive the fundamental differential equation that describes the capital per capita ($k(t) = K(t)/L(t)$) trajectory throughout time (show the intermediate steps).
2. Find the steady-state solution k^* associated to the dynamic equation found in 1.1.
3. Draw the full Solow's diagram and show the transitional dynamics for different levels of k_0 graphically.
4. Derive the Golden Rule of capital accumulation (k_G^*) (show steps); explain why inefficiency may arise (in two sentences max).

Now consider a labor augmenting production function:

$$Y(t) = K(t)^\alpha (A(t)L(t))^{1-\alpha}$$

with the following change of technological labor augmenting input:

$$\dot{A}(t) = \sigma \mu A(t)$$

5. How does the steady state stock of physical capital per capita changes compared to 1.2, assuming all else constant? Show the new steady state and explain in one sentence the change.

Finally, let us define now the change of physical capital stock by:

$$\dot{K}(t) = Y(t) - C(t) - [\delta + \phi(\mu)]K(t)$$

Where $\phi(\mu)$ the obsolescence rate. The central assumption is that a higher innovation rate (μ) leads to a faster obsolescence of existing capital: $\phi'(\mu) > 0$. population still grows at rate n . And savings rate is constant and equal to s ($0 < s < 1$). Technology grows at the same rate as in 1.5.

6. Derive the new fundamental differential equation in per effective capita of labor terms ($\hat{k} = K(t)/A(t)L(t)$)
7. Compute the new steady state solution for $\hat{k}^*$